JAVA EDITION · 30 DAYS · 2-5 HRS/DAY

Data Structures to System Design

A day-by-day plan to crack Google, Microsoft and Amazon interviews, with a live Java demo every day and a resume-grade distributed systems project.

30	162	30	20
days	LeetCode problems	live Java demos	coding patterns

Prepared for **Shubham** · Day 1: Mon 02 Nov 2026 · Day 30: Tue 01 Dec

Version 2 · updated 01 Nov 2026 (added matrices, cyclic sort, math & number theory, randomised algorithms, core Java/JVM, API design & security, distributed transactions, 20-pattern table, Java cheat sheet, interview loops)

FLAGSHIP PROJECT

NexusKV: A distributed, fault-tolerant key-value store built from scratch in Java

It is built out of the exact data structures interviewers ask about (skip list, hash map, LRU, heap, Bloom filter, consistent hashing ring) and the exact system-design topics they ask about (LSM storage, WAL, sharding, replication, leader election). Every interview question about your resume becomes a chance to show depth.

How this plan works

- Every day has: **concepts** to learn, a **live Java demo** you type and run, **LeetCode problems** (easy to hard, linked), a **NexusKV project task** on build days, and **sticky notes** to revise.
- Each topic is taught one at a time in chat: tell me *'Start Day N'* and I teach it in depth with full Java code, dry runs and interview tips, then update the sticky-notes board.
- On short (2-hour) days, do the concepts, demo and the first 3 problems. Catch up the rest on weekends.
- Rule of 25: if stuck on a problem for 25 minutes, read the hint/solution, then re-solve it from scratch 2 days later.
- Everything is in Java 21 (records, var, switch expressions, streams where it helps; plain loops in interview code).

Daily time split

2-hour day

- 40 min: concept lesson + notes
- 20 min: live Java demo (type it yourself)
- 50 min: 2-3 core problems
- 10 min: sticky-note revision

5-hour day

- 60 min: concept lesson + notes
- 30 min: live Java demo + variations
- 2 h: all listed problems (timer: 25 min each)
- 60 min: NexusKV project task
- 30 min: revise sticky notes + re-solve 1 old problem

The 30 days at a glance

Day	Date	Topic	Build
1	Mon 02 Nov	Java for DSA + Complexity Analysis	■
2	Tue 03 Nov	Arrays & Strings: Two Pointers, Sliding Window, Prefix Sums	
3	Wed 04 Nov	Hashing & HashMap Internals	
4	Thu 05 Nov	Recursion & Backtracking	
5	Fri 06 Nov	Sorting & Binary Search	
6	Sat 07 Nov	Linked Lists	■
7	Sun 08 Nov	Stacks, Queues, Deques + Week 1 Review	■
8	Mon 09 Nov	Binary Trees	
9	Tue 10 Nov	BST & Balanced Trees (AVL, Red-Black, B-Tree)	
10	Wed 11 Nov	Heaps & Priority Queues	■
11	Thu 12 Nov	Tries & String Algorithms	■
12	Fri 13 Nov	Graphs I: Representation, BFS, DFS, Topological Sort	
13	Sat 14 Nov	Graphs II: Shortest Paths, MST, Union-Find	
14	Sun 15 Nov	Week 2 Review + NexusKV Storage Engine	■
15	Mon 16 Nov	Greedy & Intervals	
16	Tue 17 Nov	Dynamic Programming I: 1D & Knapsack	
17	Wed 18 Nov	Dynamic Programming II: 2D, Strings, Grids	
18	Thu 19 Nov	DP III (Interval, Tree, Bitmask) + Bit Manipulation	■
19	Fri 20 Nov	Range Query Structures: Segment Tree, Fenwick Tree, Sparse Table	
20	Sat 21 Nov	Advanced Graphs + Union of Patterns	

Day	Date	Topic	Build
21	Sun 22 Nov	Week 3 Review + Mock Contest	■
22	Mon 23 Nov	OOP, SOLID & Design Patterns (Low-Level Design I)	■
23	Tue 24 Nov	Java Concurrency + LLD Case Studies	■
24	Wed 25 Nov	System Design Fundamentals (HLD I)	
25	Thu 26 Nov	Databases, Storage & Data Distribution (HLD II)	■
26	Fri 27 Nov	Distributed Systems & Microservices (HLD III)	■
27	Sat 28 Nov	HLD Case Studies	
28	Sun 29 Nov	NexusKV: Production Polish	■
29	Mon 30 Nov	Resume, Behavioural & Company-Specific Prep	■
30	Tue 01 Dec	Full Mock Interview Day + Final Revision	

WEEK 1

Foundations & Linear Structures

Java for DSA, complexity, arrays, hashing, recursion, sorting, linked lists, stacks/queues

DAY 1 · MON 02 NOV

Java for DSA + Complexity Analysis

Concepts

- Big-O, Big-Omega, Big-Theta; best/avg/worst case; amortized analysis
- Time vs space; counting operations; recursion-tree & Master theorem
- JVM memory: stack vs heap, references, primitives vs wrappers, autoboxing cost
- Collections Framework map: List / Set / Map / Queue / Deque and when to use each
- Fast I/O (BufferedReader, StringBuilder), Comparable vs Comparator, generics
- Core Java interview set: String/StringBuilder/StringBuffer, immutability, String pool, final/finally/finalize
- JVM & GC: heap generations, G1/ZGC basics, OutOfMemoryError vs StackOverflowError
- Java 8-21 features: lambdas, streams, Optional, records, sealed classes, var; checked vs unchecked exceptions

How to approach questions

1. Before coding, state input size n and the target complexity: n <= 10^5 needs O(n log n) or better.
2. Pick the collection by operation: lookup -> HashMap/HashSet, ordered -> TreeMap, FIFO -> ArrayDeque.
3. Finish every answer by stating time and space complexity.

LIVE JAVA DEMO

Benchmark ArrayList vs LinkedList vs ArrayDeque insert/get with System.nanoTime(); watch O(1) vs O(n) appear in real numbers.

Problems

Two Sum	Easy
Contains Duplicate	Easy
Fizz Buzz	Easy
Richest Customer Wealth	Easy
Two Sum II - Input Array Is Sorted	Medium

NexusKV build: Create the GitHub repo + Maven multi-module skeleton for NexusKV; set up JUnit 5.

STICKY NOTES

• Drop constants and lower terms: O(2n + 5) = O(n)
• Amortized O(1): ArrayList grows 1.5x when full, so n adds still cost O(n) total
• Integer caching only covers -128..127: compare wrappers with equals(), never ==
• Interviewers expect you to state time AND space before coding
• StackOverflowError = recursion too deep; raise -Xss or go iterative

DAY 2 · TUE 03 NOV

Arrays & Strings: Two Pointers, Sliding Window, Prefix Sums

Concepts

- Static vs dynamic arrays; 2D arrays; String immutability & String pool
- Two pointers (opposite ends, same direction, fast/slow)
- Sliding window: fixed size and variable size with a HashMap of counts
- Prefix sum & difference array; Kadane's algorithm
- StringBuilder, char[] frequency tables (int[26])
- Matrix traversal: spiral, rotate 90 degrees in place (transpose + reverse), set-zeroes with O(1) space
- Cyclic sort pattern for numbers in range 1..n (missing / duplicate numbers)

How to approach questions

1. Ask: contiguous? -> sliding window or prefix sums. Sorted / pairs? -> two pointers.

2. Try the O(n^2) brute force out loud, then remove the inner loop with a pointer, window or map.
3. Matrices: walk with boundaries (top, bottom, left, right) and shrink them; for in-place, use the first row/column as markers.
4. Numbers in 1..n -> cyclic sort or index marking.

LIVE JAVA DEMO

Animate a sliding window in the console: print window bounds and running sum at every step for 'max sum subarray of size k'.

Problems

Best Time to Buy and Sell Stock	Easy
Container With Most Water	Medium
3Sum	Medium
Longest Substring Without Repeating Characters	Medium
Maximum Subarray	Medium
Product of Array Except Self	Medium
Minimum Window Substring	Hard
Trapping Rain Water	Hard
Spiral Matrix	Medium
Rotate Image	Medium
Set Matrix Zeroes	Medium
Missing Number	Easy
Find All Duplicates in an Array	Medium
First Missing Positive	Hard

STICKY NOTES

• Sorted array + pair/triplet target = two pointers
• 'Longest/shortest substring with condition' = variable sliding window
• Subarray sum = k: prefix sum + HashMap
• Kadane: cur = max(x, cur + x); best = max(best, cur)
• Numbers in 1..n => cyclic sort: put x at index x-1, then scan

DAY 3 · WED 04 NOV

Hashing & HashMap Internals

Concepts

- Hash functions, load factor, rehashing, collision handling (chaining vs open addressing)
- Java HashMap internals: table of buckets, treeification to red-black tree at 8 nodes
- equals()/hashCode() contract; why mutable keys break maps
- HashSet, LinkedHashMap (insertion/access order), TreeMap (sorted, O(log n))
- Frequency counting, grouping by canonical key, rolling hash intro

How to approach questions

1. Whenever you need 'have I seen this?' or 'how many times?', reach for a HashMap/HashSet.
2. For grouping, design a canonical key (sorted string, count signature).
3. If you need order as well, consider LinkedHashMap or TreeMap.

LIVE JAVA DEMO

Implement MyHashMap from scratch with chaining and resize; then break it with a bad hashCode() and measure the slowdown.

Problems

Valid Anagram	Easy

- Group Anagrams — Medium
- Top K Frequent Elements — Medium
- Longest Consecutive Sequence — Medium
- Subarray Sum Equals K — Medium
- Design HashMap — Easy

STICKY NOTES

• Average O(1), worst O(n) (O(log n) after Java 8 treeify)
• Override equals() => must override hashCode()
• Default load factor 0.75; capacity always a power of 2
• Group by key: sorted string or int[26] signature

DAY 4 · THU 05 NOV

Recursion & Backtracking

Concepts

- Call stack, base case, recursion tree, tail recursion (Java has no TCO)
- Subsets, permutations, combinations templates
- Backtracking: choose -> explore -> un-choose; pruning
- N-Queens, Sudoku, word search on grids
- Complexity of exponential searches (2^n, n!)

How to approach questions

1. Draw the recursion tree for a tiny input first.
2. Define what one call is responsible for, then the base case.
3. Use the choose / explore / un-choose template and prune invalid branches early.

LIVE JAVA DEMO

Print the recursion tree with indentation for permutations of 'ABC', then add pruning and count saved calls.

Problems

- Subsets — Medium
- Permutations — Medium
- Combination Sum — Medium
- Generate Parentheses — Medium
- Word Search — Medium
- N-Queens — Hard

STICKY NOTES

• One template solves subsets/permutations/combinations: only the loop start and 'used' check change
• Always undo the choice after the recursive call
• Copy the path when you add it: new ArrayList<>(path)
• Prune early to cut whole subtrees

DAY 5 · FRI 06 NOV

Sorting & Binary Search

Concepts

- Bubble/Selection/Insertion (why O(n^2)); Merge sort (stable, O(n log n)); Quick sort (pivot, partition)
- Counting sort, radix sort, bucket sort; stability
- Arrays.sort: dual-pivot quicksort for primitives, TimSort for objects
- Binary search templates: exact, lower bound, upper bound
- Binary search on the answer (min capacity, Koko bananas), rotated arrays
- Divide and conquer: count inversions with merge sort, fast power x^n in O(log n)

How to approach questions

1. Sorted input or a monotonic yes/no condition -> binary search.
2. Write the check function first, then the lower-bound loop.
3. Kth largest/smallest -> heap (O(n log k)) or quickselect (average O(n)).

LIVE JAVA DEMO

Visual merge sort that prints each split and merge level; then race it against Arrays.sort on 10 million ints.

Problems

Binary Search	Easy
Search in Rotated Sorted Array	Medium
Find Minimum in Rotated Sorted Array	Medium
Koko Eating Bananas	Medium
Sort Colors	Medium
Kth Largest Element in an Array	Medium
Median of Two Sorted Arrays	Hard
Pow(x, n)	Medium
Search a 2D Matrix	Medium

STICKY NOTES

• mid = lo + (hi - lo) / 2 avoids overflow
• Monotonic yes/no condition over a range => binary search on answer
• Quickselect finds kth element in average O(n)
• Merge sort is the go-to for linked lists and counting inversions

DAY 6 · SAT 07 NOV

Linked Lists

Concepts

- Singly, doubly, circular lists; sentinel/dummy nodes
- Reverse (iterative + recursive), merge, find middle
- Fast & slow pointers: cycle detection (Floyd), cycle start
- Deep copy with random pointers
- LRU cache = HashMap + doubly linked list

How to approach questions

1. Add a dummy node whenever the head may change.
2. Middle / cycle / kth from end -> fast and slow pointers.
3. Draw 3-4 boxes with arrows on paper before re-pointing anything.

LIVE JAVA DEMO

Build a DoublyLinkedList with toString() that draws arrows (1 <-> 2 <-> 3), then build an LRU cache on top and show evictions live.

Problems

Reverse Linked List	Easy
Merge Two Sorted Lists	Easy
Linked List Cycle	Easy
Remove Nth Node From End of List	Medium
Reorder List	Medium
Copy List with Random Pointer	Medium
LRU Cache	Medium
Reverse Nodes in k-Group	Hard

NexusKV build: NexusKV: implement the LRU read cache (HashMap + DLL) with unit tests.

STICKY NOTES

• Dummy head removes edge cases for deleting the first node
• Fast/slow: fast moves 2, slow 1; they meet inside a cycle
• Reverse: prev=null; while(cur){next=cur.next; cur.next=prev; prev=cur; cur=next}
• LRU get/put are both O(1)

DAY 7 · SUN 08 NOV

Stacks, Queues, Deques + Week 1 Review

Concepts

- Stack (use ArrayDeque, not Stack), queue, circular queue, deque
- Monotonic stack: next greater element, histogram
- Monotonic deque: sliding window maximum
- Expression evaluation, valid parentheses, min stack
- Week 1 review: 10 mixed problems timed at 25 min each
- Implement queue using stacks and stack using queues (asked as warm-ups)

How to approach questions

1. Matching brackets or 'undo' behaviour -> stack.
2. 'Next greater/smaller' or 'span' -> monotonic stack of indices.
3. Window maximum -> monotonic deque. Level-by-level -> queue.

LIVE JAVA DEMO

Monotonic stack trace for 'Daily Temperatures': print the stack after every push and pop.

Problems

Valid Parentheses	Easy
Min Stack	Medium
Daily Temperatures	Medium
Evaluate Reverse Polish Notation	Medium
Sliding Window Maximum	Hard
Largest Rectangle in Histogram	Hard
Implement Queue using Stacks	Easy
Design Circular Queue	Medium

NexusKV build: NexusKV: in-memory store with GET/PUT/DELETE and a skip-list memtable (ordered keys).

STICKY NOTES

• Deque st = new ArrayDeque<>(); is the modern stack
• 'Next greater/smaller' = monotonic stack, O(n)
• Sliding window max = deque of indices, decreasing values
• Skip list: O(log n) expected search with simple code (used in Redis & LevelDB)

WEEK 2

Trees, Heaps, Tries & Graphs

Hierarchical and network structures plus the classic graph algorithms

DAY 8 · MON 09 NOV

Binary Trees

Concepts

- Terminology: height, depth, full/complete/perfect/balanced
- DFS traversals: pre/in/post order (recursive + iterative)
- BFS level-order with a queue; zigzag; right-side view
- Return-value recursion: height, diameter, max path sum
- Serialize/deserialize, LCA, construct from traversals

How to approach questions

1. Decide what each call returns for its subtree and whether you also need a global answer.
2. Level-based wording -> BFS with queue size per level.
3. Path problems: compute best downward path, update global with left + right through the node.

LIVE JAVA DEMO

Pretty-print a binary tree sideways in the console and show all 4 traversal orders for the same tree.

Problems

Maximum Depth of Binary Tree	Easy
Invert Binary Tree	Easy
Diameter of Binary Tree	Easy
Binary Tree Level Order Traversal	Medium
Binary Tree Right Side View	Medium
Lowest Common Ancestor of a Binary Tree	Medium
Binary Tree Maximum Path Sum	Hard
Serialize and Deserialize Binary Tree	Hard

STICKY NOTES

• Most tree problems: solve for left, solve for right, combine
• Level-order: record queue.size() before the inner loop
• Global answer (diameter, path sum) + returned value (height) is a common pair
• Iterative inorder uses an explicit stack

DAY 9 · TUE 10 NOV

BST & Balanced Trees (AVL, Red-Black, B-Tree)

Concepts

- BST insert/search/delete (3 deletion cases)
- Validate BST with min/max bounds; inorder = sorted
- AVL rotations (LL, RR, LR, RL); Red-Black properties (concept level)
- TreeMap/TreeSet API: floorKey, ceilingKey, headMap, subMap
- B-Tree / B+Tree: why databases index with them

How to approach questions

1. Use the BST property to discard half the tree at each step.
2. Inorder traversal gives sorted order: kth smallest, validate, convert.
3. In Java, floor/ceiling queries -> TreeMap/TreeSet instead of writing a BST.

LIVE JAVA DEMO

Insert 1..7 into a plain BST (degenerates to a list) vs an AVL tree; print heights to show O(n) vs O(log n).

Problems

Validate Binary Search Tree	Medium
Kth Smallest Element in a BST	Medium
Lowest Common Ancestor of a Binary Search Tree	Medium
Delete Node in a BST	Medium
Convert Sorted Array to Binary Search Tree	Easy
My Calendar I	Medium

STICKY NOTES

• TreeMap is a red-black tree: all ops O(log n)
• floorKey/ceilingKey solve 'closest value' problems instantly
• Validate BST with (min,max) bounds, not just parent comparison
• B+Trees keep data in leaves linked for fast range scans

DAY 10 · WED 11 NOV

Heaps & Priority Queues

Concepts

- Binary heap as an array: parent (i-1)/2, children 2i+1, 2i+2
- Sift up/down; heapify in O(n); heap sort
- PriorityQueue with comparators; min-heap vs max-heap
- Top-K pattern, K-way merge, two heaps for running median
- Scheduling problems (task scheduler, meeting rooms)

How to approach questions

1. K largest -> min-heap of size K; K smallest/closest -> max-heap of size K.
2. Merging sorted sources -> heap of the current head of each source.
3. Median or two halves -> two heaps; scheduling -> heap of next available time.

LIVE JAVA DEMO

Implement MinHeap from scratch and print the array after every insert/poll, side by side with its tree shape.

Problems

Kth Largest Element in a Stream	Easy
Last Stone Weight	Easy
K Closest Points to Origin	Medium
Task Scheduler	Medium
Merge k Sorted Lists	Hard
Find Median from Data Stream	Hard

NexusKV build: NexusKV: K-way merge iterator (heap) to merge sorted runs.

STICKY NOTES

• Top K largest => min-heap of size K (O(n log K))
• Max-heap: new PriorityQueue<>(Collections.reverseOrder())
• Running median: max-heap low half + min-heap high half
• Building a heap from an array is O(n), not O(n log n)

DAY 11 · THU 12 NOV

Tries & String Algorithms

Concepts

- Trie node (children[26] or HashMap), insert/search/startsWith
- Autocomplete, word dictionary with wildcards, word search II
- Pattern matching: KMP (LPS array), Rabin-Karp (rolling hash), Z-algorithm
- Palindromes: expand-around-center, Manacher (overview)
- Bloom filter: probabilistic membership with k hash functions

How to approach questions

1. Many words + prefix queries -> trie.
2. Find a pattern in text -> KMP or Rabin-Karp; for interviews indexOf is fine unless asked to implement.
3. Palindromes -> expand around each centre (2n - 1 centres).

LIVE JAVA DEMO

Build an autocomplete engine over 10,000 words with a Trie; type a prefix and get top suggestions in microseconds.

Problems

■ Implement Trie (Prefix Tree)	Medium
■ Design Add and Search Words Data Structure	Medium
■ Longest Palindromic Substring	Medium
■ Find the Index of the First Occurrence in a String	Easy
■ Word Search II	Hard

NexusKV build: NexusKV: Bloom filter per data file to skip disk reads for missing keys.

STICKY NOTES

• Trie search is O(length of word), independent of dictionary size
• KMP never moves the text pointer backwards: O(n + m)
• Bloom filter: false positives possible, false negatives never
• Rabin-Karp: compare hashes first, characters only on a hash match

DAY 12 · FRI 13 NOV

Graphs I: Representation, BFS, DFS, Topological Sort

Concepts

- Adjacency list vs matrix vs edge list; directed/undirected/weighted
- BFS (shortest path in unweighted graphs), DFS (connectivity, cycles)
- Grid as a graph: islands, flood fill, multi-source BFS
- Cycle detection: undirected (parent) and directed (colours)
- Topological sort: Kahn's (BFS, in-degree) and DFS post-order
- Multi-source BFS on grids and implicit graphs (word ladder, open the lock)

How to approach questions

1. First identify nodes and edges; grids are graphs with 4 neighbours.
2. Fewest steps -> BFS. Reachability / components -> DFS or BFS. Ordering with prerequisites -> topological sort.
3. Mark visited when you enqueue.

LIVE JAVA DEMO

Grid BFS that prints the board after every wave for 'Rotting Oranges', so you watch the multi-source BFS spread.

Problems

■ Number of Islands	Medium

- Clone Graph — Medium
- Rotting Oranges — Medium
- Pacific Atlantic Water Flow — Medium
- Course Schedule — Medium
- Course Schedule II — Medium
- Word Ladder — Hard
- Open the Lock — Medium
- Surrounded Regions — Medium

STICKY NOTES

• Unweighted shortest path = BFS
• Dependencies/ordering = topological sort; if fewer than n nodes come out, there is a cycle
• Mark visited when you ENQUEUE, not when you dequeue
• Grid neighbours: int[][] dirs = {{1,0},{-1,0},{0,1},{0,-1}}

DAY 13 · SAT 14 NOV

Graphs II: Shortest Paths, MST, Union-Find

Concepts

- Dijkstra with PriorityQueue (non-negative weights)
- Bellman-Ford (negative edges, k-stops), Floyd-Warshall (all pairs)
- 0-1 BFS with deque
- Union-Find with path compression + union by rank
- Minimum spanning tree: Kruskal (sort + DSU), Prim (heap)

How to approach questions

1. Weighted shortest path, non-negative -> Dijkstra. Negative edges or 'at most k stops' -> Bellman-Ford.
2. Connecting all nodes at minimum cost -> MST (Kruskal with DSU).
3. Dynamic connectivity or cycle in undirected graph -> Union-Find.

LIVE JAVA DEMO

Dijkstra on a small city map of Chennai areas; print the PQ and the distance table after every relaxation.

Problems

- Network Delay Time — Medium
- Cheapest Flights Within K Stops — Medium
- Number of Provinces — Medium
- Redundant Connection — Medium
- Min Cost to Connect All Points — Medium
- Path With Minimum Effort — Medium
- Swim in Rising Water — Hard

STICKY NOTES

• Dijkstra: skip stale PQ entries (if d > dist[u]) continue;
• Negative weights => Bellman-Ford, never Dijkstra
• DSU with both optimisations is almost O(1) per op (inverse Ackermann)
• Kruskal = sort edges + DSU; Prim = PQ from a start node

DAY 14 · SUN 15 NOV

Week 2 Review + NexusKV Storage Engine

Concepts

- Timed mock: 2 medium + 1 hard in 90 minutes
- Revise all sticky notes from Days 8-13
- LSM-tree design: memtable -> immutable memtable -> SSTable files
- Write-ahead log (WAL) for durability and crash recovery
- Java NIO: FileChannel, ByteBuffer, fsync (force)

How to approach questions

1. For design-a-data-structure questions, list each operation and its required complexity first.
2. Combine structures: HashMap for lookup + list/heap/tree for order.
3. Write the API signatures, then implement one method at a time with tests.

LIVE JAVA DEMO

Kill the NexusKV process mid-write, restart it, and watch the WAL replay recover every key.

Problems

- Time Based Key-Value Store — Medium
- Design Twitter — Medium
- Snapshot Array — Medium

NexusKV build: NexusKV: WAL + flush memtable to sorted SSTable files with an index block and Bloom filter.

STICKY NOTES

• LSM trees turn random writes into sequential writes
• WAL first, memtable second: that order is what makes writes durable
• SSTables are immutable; deletes are tombstones
• Read path: cache -> memtable -> SSTables newest to oldest (Bloom filter first)

WEEK 3

Advanced Algorithms

Greedy, the four flavours of DP, bit tricks, segment/Fenwick trees, advanced graphs

DAY 15 · MON 16 NOV

Greedy & Intervals

Concepts

- Greedy choice property and exchange argument (how to prove greedy is correct)
- Interval patterns: merge, insert, non-overlapping, meeting rooms
- Jump game, gas station, partition labels
- Sweep line with sorted events
- When greedy fails (coin change with arbitrary coins)

How to approach questions

1. Intervals: sort by start to merge, by end to maximise non-overlapping.
2. Try a counter-example before trusting greedy.
3. Reachability like Jump Game: track the farthest index you can reach.

LIVE JAVA DEMO

Sweep line over meeting intervals: print active meetings at every event to find minimum rooms.

Problems

Merge Intervals	Medium
Insert Interval	Medium
Non-overlapping Intervals	Medium
Jump Game	Medium
Gas Station	Medium
Partition Labels	Medium
Minimum Number of Arrows to Burst Balloons	Medium

STICKY NOTES

• Intervals: sort by start (merging) or by end (max non-overlapping)
• Meeting rooms = min-heap of end times, or sweep line
• Greedy needs a proof; if you can find a counter-example, use DP
• Sort with Arrays.sort(a, (x, y) -> Integer.compare(x[0], y[0]))

DAY 16 · TUE 17 NOV

Dynamic Programming I: 1D & Knapsack

Concepts

- Overlapping subproblems + optimal substructure
- Top-down memoization vs bottom-up tabulation vs space optimisation
- The 5 steps: state, transition, base case, order, answer
- 1D DP: climbing stairs, house robber, decode ways
- 0/1 knapsack, unbounded knapsack, subset sum, coin change

How to approach questions

1. Signals for DP: 'number of ways', 'min/max cost', 'can you reach', with overlapping choices.
2. Write recursion -> memo -> table -> space-optimised.
3. Knapsack-type: for each item decide take / skip; mind the loop direction.

LIVE JAVA DEMO

Fibonacci three ways (naive, memo, table) with call counters: 1.6 billion calls vs 90 for n=45.

Problems

Climbing Stairs	Easy
House Robber	Medium
House Robber II	Medium
Decode Ways	Medium
Coin Change	Medium
Partition Equal Subset Sum	Medium
Longest Increasing Subsequence	Medium
Word Break	Medium

STICKY NOTES

• Write the recursion first, then add memo, then convert to a table
• 0/1 knapsack 1D: loop capacity BACKWARDS
• Unbounded knapsack 1D: loop capacity FORWARDS
• LIS in O(n log n): patience sorting with binary search

DAY 17 · WED 18 NOV

Dynamic Programming II: 2D, Strings, Grids

Concepts

- Grid paths with obstacles; minimum path sum
- LCS, edit distance, distinct subsequences
- Palindromic substrings/subsequences
- Stock problems as state machines
- Rolling arrays to reduce 2D space to 1D

How to approach questions

1. Two strings -> dp[i][j] over prefixes; one grid -> dp[r][c] from top/left.
2. Fill a tiny table by hand to find the transition.
3. State machine for stock problems: states = holding / not holding / cooldown.

LIVE JAVA DEMO

Print the full edit-distance table for 'horse' -> 'ros' and trace back the actual edit operations.

Problems

Unique Paths	Medium
Minimum Path Sum	Medium
Longest Common Subsequence	Medium
Edit Distance	Medium
Palindromic Substrings	Medium
Best Time to Buy and Sell Stock with Cooldown	Medium
Distinct Subsequences	Hard
Regular Expression Matching	Hard

STICKY NOTES

• Two strings => dp[i][j] over prefixes of length i and j
• Edit distance: 1 + min(insert, delete, replace) when chars differ
• Stock problems: one state per (day, holding?, transactions left)
• Draw the table by hand for a tiny input before coding

DAY 18 · THU 19 NOV

DP III (Interval, Tree, Bitmask) + Bit Manipulation

Concepts

- Interval DP: burst balloons, matrix chain
- DP on trees: house robber III, max path
- Bitmask DP: travelling salesman, assign tasks
- Bit tricks: x & (x-1), x & -x, XOR properties, masks
- Counting bits, single number, subsets via bitmask
- Math & number theory: GCD/LCM (Euclid), Sieve of Eratosthenes, modular arithmetic (1e9+7), modular inverse, nCr
- Randomised algorithms (Google favourite): reservoir sampling, weighted random pick, Fisher-Yates shuffle

How to approach questions

1. Interval DP: dp[l][r] over lengths from small to large; try every split point.
2. n <= 20 and 'visit all' -> bitmask DP over subsets.
3. Math: use long and % MOD after every multiply; gcd via Euclid; primes via sieve.
4. Random: reservoir sampling for streams, prefix sums + binary search for weighted picks.

LIVE JAVA DEMO

Enumerate all subsets of {a,b,c,d} with bitmasks and print each mask in binary next to its subset.

Problems

Single Number	Easy
Counting Bits	Easy
Number of 1 Bits	Easy
House Robber III	Medium
Target Sum	Medium
Burst Balloons	Hard
Shortest Path Visiting All Nodes	Hard
Count Primes	Medium
Linked List Random Node	Medium
Random Pick with Weight	Medium
Insert Delete GetRandom O(1)	Medium

NexusKV build: NexusKV: background compaction (merge SSTables with the heap iterator, drop tombstones).

STICKY NOTES

- a ^ a = 0 and a ^ 0 = a: XOR finds the single element
- x & (x - 1) clears the lowest set bit
- Interval DP: loop by LENGTH of interval, outermost
- n <= 20 with 'visit all' => bitmask DP
- Take % MOD after every multiply; use long for the intermediate

DAY 19 · FRI 20 NOV

Range Query Structures: Segment Tree, Fenwick Tree, Sparse Table

Concepts

- Segment tree: build, point update, range query, O(log n)
- Lazy propagation for range updates
- Fenwick tree (Binary Indexed Tree): prefix sums with i & -i
- Sparse table for O(1) range min on static arrays
- Ordered statistics; coordinate compression

How to approach questions

1. Static array, many range queries -> prefix sums or sparse table.
2. Updates + range queries -> Fenwick tree (sums) or segment tree (min/max/sum).
3. Counting smaller elements -> Fenwick over compressed values, or merge sort.

LIVE JAVA DEMO

Segment tree over [2,1,5,3,4]: print the tree array after build and after each update, and trace a range-sum query.

Problems

Range Sum Query - Mutable	Medium
Count of Smaller Numbers After Self	Hard
Range Sum Query 2D - Immutable	Medium
My Calendar III	Hard
Count of Range Sum	Hard

STICKY NOTES

- Static array => prefix sums; updates too => Fenwick/segment tree
- Segment tree array size: 4n is always safe
- Fenwick: update i += i & -i, query i -= i & -i (1-indexed)
- Lazy propagation = defer updates until a query needs them

DAY 20 · SAT 21 NOV

Advanced Graphs + Union of Patterns

Concepts

- Strongly connected components: Kosaraju & Tarjan
- Bridges and articulation points (low-link values)
- Eulerian path (Hierholzer), bipartite check
- A* search; max-flow intro (Ford-Fulkerson / Edmonds-Karp)
- The 15 coding patterns cheat sheet: map any question to a pattern in 2 minutes
- LCA with binary lifting (O(log n) per query) and Euler tour basics

How to approach questions

1. Critical edges/nodes -> Tarjan low-link values.
2. Two-colouring -> BFS with colours; odd cycle means not bipartite.
3. Use every edge exactly once -> Eulerian path (Hierholzer).

LIVE JAVA DEMO

Tarjan's algorithm on a sample network: print discovery and low values and highlight the critical connections.

Problems

Is Graph Bipartite?	Medium
Critical Connections in a Network	Hard
Reconstruct Itinerary	Hard
Verifying an Alien Dictionary	Easy

- Bus Routes — Hard
- Kth Ancestor of a Tree Node — Hard

STICKY NOTES

• Bridge (u,v) when low[v] > disc[u]
• Bipartite = 2-colourable = no odd cycle
• Eulerian path: at most 2 nodes with odd degree
• Pattern first, code second: say the pattern out loud in interviews

DAY 21 · SUN 22 NOV

Week 3 Review + Mock Contest

Concepts

- Virtual LeetCode contest (4 problems, 90 minutes)
- Re-solve every problem you marked 'needed hint' this week
- Revise DP sticky notes; write the state+transition for 10 DP problems without coding
- Java NIO networking: ServerSocketChannel + Selector
- Binary protocol design (length-prefixed messages)

How to approach questions

1. Contest order: read all, solve the easiest first, cap each problem at 20 minutes before moving on.
2. For every miss, write the pattern you failed to spot in your mistake log.

LIVE JAVA DEMO

Connect to NexusKV over TCP from two terminals at once and run SET/GET commands, like redis-cli.

Problems

- LFU Cache — Hard
- Number of Recent Calls — Easy
- Max Points on a Line — Hard

NexusKV build: NexusKV: non-blocking TCP server (Java NIO) with a simple text protocol + client CLI.

STICKY NOTES

• Contest rhythm: read all 4, solve easiest first, never stall more than 20 minutes
• For DP, the state definition is 80% of the solution
• One Selector thread can serve thousands of sockets
• LFU = HashMap + HashMap + minFreq

WEEK 4

System Design: LLD, Concurrency & HLD

OOP/SOLID/patterns, Java concurrency, HLD building blocks, distributed systems, mocks

DAY 22 · MON 23 NOV

OOP, SOLID & Design Patterns (Low-Level Design I)

Concepts

- Encapsulation, abstraction, inheritance vs composition, polymorphism
- SOLID with Java examples (and how each one is violated)
- Creational: Singleton (enum/double-checked), Factory, Builder
- Structural: Adapter, Decorator, Proxy, Facade
- Behavioural: Strategy, Observer, Command, State, Chain of Responsibility

How to approach questions

1. LLD round: clarify requirements and actors, list entities, define relationships.
2. Draw the class diagram, then code the core classes and one key flow.
3. Point out where patterns give extensibility (Strategy for pricing, Observer for notifications).

LIVE JAVA DEMO

Refactor a 200-line 'god class' payment processor into Strategy + Factory and show how adding UPI becomes a one-class change.

Problems

- ■ Design Parking System — Easy
- ■ Design Browser History — Medium
- ■ Design Underground System — Medium

NexusKV build: NexusKV: refactor command handling into Command pattern + Strategy for eviction policies.

STICKY NOTES

• Favour composition over inheritance
• Open/Closed: add behaviour by adding classes, not editing switch statements
• Enum singleton is serialization- and reflection-safe
• LLD interview flow: requirements -> entities -> relations -> class diagram -> code -> extensibility

DAY 23 · TUE 24 NOV

Java Concurrency + LLD Case Studies

Concepts

- Threads, Runnable/Callable, ExecutorService, CompletableFuture
- synchronized, volatile, happens-before, ReentrantLock, ReadWriteLock
- ConcurrentHashMap, AtomicInteger, BlockingQueue, CountDownLatch
- Deadlock (4 conditions), producer-consumer, thread-safe singleton
- LLD case studies: Parking Lot, Elevator, BookMyShow, Splitwise, Rate Limiter class

How to approach questions

1. Identify shared mutable state first; protect it or make it immutable.
2. Prefer high-level tools (ConcurrentHashMap, ExecutorService, BlockingQueue) over raw locks.
3. Talk about deadlock, starvation and how you would test concurrency.

LIVE JAVA DEMO

Race condition live: 1,000 threads incrementing a counter (wrong answer) -> fix with synchronized -> AtomicInteger -> LongAdder, with timings.

Problems

- ■ Print in Order — Easy

- Print FooBar Alternately — Medium
- Building H2O — Medium

NexusKV build: NexusKV: thread-safe store (ReadWriteLock / ConcurrentSkipListMap) + JMH benchmark of ops/sec.

STICKY NOTES

• volatile = visibility, NOT atomicity
• Prefer ExecutorService over new Thread()
• ConcurrentHashMap locks per bin, never the whole map
• Deadlock needs: mutual exclusion, hold & wait, no preemption, circular wait

DAY 24 · WED 25 NOV

System Design Fundamentals (HLD I)

Concepts

- The interview framework: requirements -> estimates -> API -> data model -> HLD -> deep dives -> bottlenecks
- Back-of-envelope estimation: QPS, storage, bandwidth
- Vertical vs horizontal scaling, stateless services, load balancers (L4/L7, algorithms)
- Caching: cache-aside, write-through, write-back, eviction, CDN
- CAP & PACELC, consistency models (strong, eventual, read-your-writes)
- API design: REST resources, status codes, pagination (cursor vs offset), versioning, idempotent methods
- Security basics: authentication vs authorisation, sessions vs JWT, OAuth 2.0 flow, HTTPS/TLS
- Distributed unique IDs: UUID vs DB sequence vs Snowflake (timestamp + machine + sequence)

How to approach questions

1. Always open with requirements and back-of-envelope numbers.
2. Draw client -> load balancer -> stateless services -> cache -> database.
3. For each component say why it is there and its trade-off.

LIVE JAVA DEMO

Run 3 Spring Boot instances behind an Nginx load balancer in Docker Compose; kill one and watch traffic shift.

STICKY NOTES

• 1 day = 86,400 s; 1M requests/day = about 12 QPS
• Cache-aside is the default; always mention invalidation and TTL
• Stateless app servers scale horizontally; state lives in DB/cache
• CAP: under a partition you choose consistency or availability
• Cursor pagination stays correct when rows are inserted; offset pagination does not

DAY 25 · THU 26 NOV

Databases, Storage & Data Distribution (HLD II)

Concepts

- SQL vs NoSQL (key-value, document, wide-column, graph) and when to pick each
- Indexes: B+Tree vs LSM-tree; ACID, isolation levels
- Replication: leader-follower, multi-leader, leaderless (quorum R + W > N)
- Partitioning/sharding: range vs hash; consistent hashing with virtual nodes
- Hot keys, rebalancing, secondary indexes
- SQL interview essentials: joins, GROUP BY/HAVING, window functions, normalisation vs denormalisation
- Blob/object storage (S3-style) vs block storage; storing media metadata separately

How to approach questions

1. Choose storage from access patterns, not habit.
2. State the sharding key and how you handle hot keys and rebalancing.
3. Explain replication mode and what happens when the leader fails.

LIVE JAVA DEMO

Consistent hashing ring in Java: add/remove nodes and print how many keys move (about 1/N vs almost all with mod N).

NexusKV build: NexusKV: shard keys across 3 nodes with consistent hashing (virtual nodes).

STICKY NOTES

- Consistent hashing moves only about K/N keys when a node joins or leaves
- Quorum: R + W > N gives you overlapping reads/writes
- Read-heavy => replicas + cache; write-heavy => sharding + LSM
- Always state your sharding key and why

DAY 26 · FRI 27 NOV

Distributed Systems & Microservices (HLD III)

Concepts

- Message queues & streams: Kafka partitions, consumer groups, delivery semantics
- Consensus: Raft leader election and log replication
- Rate limiting algorithms: token bucket, leaky bucket, sliding window
- Microservices: API gateway, service discovery, circuit breaker, saga, idempotency keys
- Observability: logs, metrics, tracing; REST vs gRPC vs WebSockets
- Distributed transactions: 2PC vs saga (choreography/orchestration), outbox pattern
- CQRS & event sourcing; search with an inverted index (Elasticsearch)
- Time and ordering: clock skew, vector clocks, at-least-once delivery

How to approach questions

1. Use a queue whenever work can be async or traffic is spiky.
2. Every cross-service write: discuss idempotency, retries and the outbox pattern.
3. Name your consistency choice and the failure you are guarding against.

LIVE JAVA DEMO

Token-bucket rate limiter in Java hit by a load generator; watch 429s appear exactly at the configured rate.

NexusKV build: NexusKV: leader-follower replication + heartbeat-based failover (Raft-lite leader election).

STICKY NOTES

- Exactly-once is really at-least-once + idempotent consumers
- Raft: a leader needs votes from a majority; terms prevent split brain
- Token bucket allows bursts; leaky bucket smooths them
- Circuit breaker states: closed -> open -> half-open
- Never update a DB and publish an event separately: use the outbox pattern

DAY 27 · SAT 28 NOV

HLD Case Studies

Concepts

- URL shortener (base62, key generation service)
- Rate limiter as a distributed service (Redis + Lua)
- Chat app / WhatsApp (WebSockets, fan-out, message ordering)
- News feed (push vs pull vs hybrid), YouTube/Netflix (CDN, transcoding)
- Uber/ride matching (geohash, quadtree), Google Docs (OT/CRDT overview)
- More case studies: notification system, distributed cache, typeahead/autocomplete, web crawler, Dropbox/Google Drive, ticket booking (BookMyShow)

How to approach questions

1. Run the 7-step framework; spend most time on the 1-2 hardest deep dives.
2. Use numbers to justify every scaling decision.

3. Close with bottlenecks and what changes at 10x scale.

LIVE JAVA DEMO

Whiteboard-style walkthrough of the URL shortener end to end, then build its core (base62 + store) in 40 lines of Java.

Problems

- Encode and Decode TinyURL — **Medium**
- Search Suggestions System — **Medium**

STICKY NOTES

• Every design: clarify scale first (DAU, read/write ratio, latency target)
• Celebrity problem in feeds => hybrid push/pull
• Geospatial search => geohash or quadtree
• Name trade-offs out loud; there is no single right design
• Ticket booking: lock seats with a TTL hold to avoid double booking

DAY 28 · SUN 29 NOV

NexusKV: Production Polish

Concepts

- Dockerfile + Docker Compose for a 3-node cluster
- GitHub Actions CI: build, test, JaCoCo coverage badge
- JMH benchmarks: throughput and p99 latency tables
- Spring Boot admin dashboard: node health, key counts, ring view
- Architecture diagram + design doc (ADR style)

How to approach questions

1. Measure before optimising: write the benchmark first.
2. Explain your project in the same 7-step framework you use in interviews.

LIVE JAVA DEMO

docker compose up -> 3 nodes start -> run the benchmark -> kill the leader -> watch failover on the dashboard.

NexusKV build: NexusKV: Docker, CI, benchmarks, dashboard, README with architecture diagram and numbers.

STICKY NOTES

• A project without numbers is a claim; with benchmarks it is evidence
• README order: what, why, architecture, quick start, benchmarks, design decisions
• Keep commits small and meaningful: reviewers read history
• Record a 2-minute demo GIF for the README

DAY 29 · MON 30 NOV

Resume, Behavioural & Company-Specific Prep

Concepts

- Resume bullets with impact: action + what + measurable result
- Amazon Leadership Principles: 2 STAR stories each for the top 8
- Google: Googleyness, structured problem solving, thinking aloud
- Microsoft: growth mindset, collaboration, design-heavy rounds
- Turning Hitachi Energy work into strong stories (sole-developer features, awards)
- Application strategy: referrals, LinkedIn headline + project post, OA preparation (HackerRank/Codility style), offer negotiation basics

How to approach questions

1. STAR: short situation, clear 'I' actions, measurable result.

2. Map each story to 2-3 Amazon Leadership Principles.

LIVE JAVA DEMO

Rewrite 3 weak resume bullets live into quantified, interview-ready bullets.

Problems

- LRU Cache (re-solve in 15 min) — **Medium**
- Merge k Sorted Lists (re-solve) — **Hard**
- Course Schedule II (re-solve) — **Medium**

NexusKV build: NexusKV: final review; pin repo on GitHub; add it to the resume and LinkedIn.

STICKY NOTES

• STAR: keep Situation+Task under 20% of the answer; Result must have a number
• Prepare a 'conflict', a 'failure', and a 'tight deadline' story
• Every resume line is fair game: be ready to go 3 levels deep
• Ask the interviewer a real question at the end

DAY 30 · TUE 01 DEC

Full Mock Interview Day + Final Revision

Concepts

- Mock 1: DSA (2 problems, 45 min, think aloud)
- Mock 2: LLD (design a parking lot or elevator in 45 min)
- Mock 3: HLD (design a URL shortener or chat app in 45 min)
- Mock 4: behavioural (6 questions)
- Revise all 30 days of sticky notes; build the 60-day post-plan

How to approach questions

1. Clarify -> examples -> brute force -> optimise -> code -> dry run -> complexity.
2. Think aloud the whole time; silence is the most common reason for a weak signal.

LIVE JAVA DEMO

Recorded mock interview: review the recording for silence gaps, missed edge cases and complexity statements.

Problems

- Word Ladder (timed) — **Hard**
- Longest Increasing Path in a Matrix — **Hard**
- Design a Food Rating System — **Medium**

STICKY NOTES

• Interview loop: clarify -> examples -> brute force -> optimise -> code -> test -> complexity
• Test with an edge case before saying 'done'
• Keep solving 2 problems a day after Day 30: consistency beats cramming
• You are ready when you can explain WHY a pattern applies, not just recall it

FLAGSHIP PROJECT

NexusKV: A distributed, fault-tolerant key-value store built from scratch in Java

It is built out of the exact data structures interviewers ask about (skip list, hash map, LRU, heap, Bloom filter, consistent hashing ring) and the exact system-design topics they ask about (LSM storage, WAL, sharding, replication, leader election). Every interview question about your resume becomes a chance to show depth.

Architecture

```
Client CLI → NIO TCP server → Router (consistent-hash ring) → Leader node → [WAL → Skip-list memtable → SSTables + Bloom filters → Compaction] → replicated to Follower nodes (heartbeats + leader election). Spring Boot admin dashboard reads node health and metrics.
```

Features

Storage engine	LSM-tree: skip-list memtable, WAL, immutable SSTables with index + Bloom filter, background compaction
Caching	O(1) LRU read cache (HashMap + doubly linked list), pluggable LFU strategy
Networking	Non-blocking Java NIO server, length-prefixed protocol, CLI client
Distribution	Consistent hashing with virtual nodes across 3+ nodes
Fault tolerance	Leader-follower replication, heartbeats, Raft-lite leader election, crash recovery from WAL
Engineering	Docker Compose cluster, CI pipeline, test coverage, JMH throughput & p99 latency numbers, admin dashboard

Tech stack

Java 21 · Maven · Java NIO (TCP server) · Spring Boot (admin dashboard + REST) · JUnit 5 + Mockito · JMH benchmarks · Docker + Docker Compose · GitHub Actions CI

Build timeline

- **Day 1:** Create the GitHub repo + Maven multi-module skeleton for NexusKV; set up JUnit 5.
- **Day 6:** implement the LRU read cache (HashMap + DLL) with unit tests.
- **Day 7:** in-memory store with GET/PUT/DELETE and a skip-list memtable (ordered keys).
- **Day 10:** K-way merge iterator (heap) to merge sorted runs.
- **Day 11:** Bloom filter per data file to skip disk reads for missing keys.
- **Day 14:** WAL + flush memtable to sorted SSTable files with an index block and Bloom filter.
- **Day 18:** background compaction (merge SSTables with the heap iterator, drop tombstones).
- **Day 21:** non-blocking TCP server (Java NIO) with a simple text protocol + client CLI.
- **Day 22:** refactor command handling into Command pattern + Strategy for eviction policies.
- **Day 23:** thread-safe store (ReadWriteLock / ConcurrentSkipListMap) + JMH benchmark of ops/sec.
- **Day 25:** shard keys across 3 nodes with consistent hashing (virtual nodes).
- **Day 26:** leader-follower replication + heartbeat-based failover (Raft-lite leader election).
- **Day 28:** Docker, CI, benchmarks, dashboard, README with architecture diagram and numbers.
- **Day 29:** final review; pin repo on GitHub; add it to the resume and LinkedIn.

Resume bullets (fill in your benchmark numbers)

- Built NexusKV, a distributed key-value store in Java 21 with an LSM-tree storage engine (skip-list memtable, WAL, SSTables, Bloom filters), sustaining ~XXk ops/sec at p99 < X ms in JMH benchmarks.
- Designed sharding with consistent hashing (virtual nodes) and leader-follower replication with automatic failover, recovering from node crashes with zero acknowledged-write loss.

- Implemented a non-blocking Java NIO server, O(1) LRU cache and background compaction; shipped with Docker Compose, GitHub Actions CI and 85%+ test coverage.

Lighter alternative if time runs short: a Spring Boot microservices URL shortener + distributed rate limiter (Redis, Docker, API gateway). Less impressive, but finishable in 4 weekends.

Extra topics included (beyond pure DSA)

Java Collections internals	HashMap, ArrayList, ArrayDeque, PriorityQueue, TreeMap, ConcurrentHashMap: how they work inside (asked often at Amazon/Microsoft).
Java concurrency	Day 23. Needed for LLD rounds and for the project.
15 coding patterns	Two pointers, sliding window, fast/slow, merge intervals, cyclic sort, in-place reversal, BFS, DFS, two heaps, subsets, modified binary search, top-K, K-way merge, topological sort, DP.
Low-Level Design	Days 22-23. A separate round at Amazon, Microsoft, Flipkart, Uber.
Behavioural rounds	Day 29. Amazon weights Leadership Principles heavily in every round.
Docker & CI/CD basics	Learned through the project on Day 28 (fills a gap on your profile).
Mock interviews	Days 21 and 30. Use Pramp / interviewing.io peers or record yourself.

Interview day framework

- **DSA:** clarify → examples → brute force → optimise → code → dry run → complexity
- **LLD:** requirements → entities → relationships → class diagram → code core classes → patterns & extensibility
- **HLD:** requirements → estimates → API → data model → high-level diagram → deep dives → bottlenecks & trade-offs
- **Behavioural:** STAR, with a number in every Result

Resources

- **LeetCode**: Every problem in this plan links here
- **NeetCode 150 roadmap**: Pattern-wise problem list
- **VisuAlgo**: Animations of every data structure
- **System Design Primer**: Free HLD reference
- **Designing Data-Intensive Applications**: The book behind NexusKV's design (Ch. 3, 5, 6, 9)
- **Refactoring Guru**: Design patterns with Java examples

After Day 30

- Keep 2 problems/day (one new, one re-solve) until your interviews.
- One HLD case study per week from the list on Day 27.
- Weekly LeetCode contest every Sunday morning.

SYSTEM DESIGN HANDBOOK

System design at a glance (Days 22-28)

Everything system design in one place: the answer framework, the building blocks, the case studies and low-level design. Each row names the day it is taught.

The 7-step answer framework

1. Requirements	Functional (what it does) and non-functional (scale, latency, availability, consistency). Agree on scope in 5 minutes.
2. Estimates	DAU, read/write QPS, peak = 2-3x average, storage per year, bandwidth.
3. API	Endpoints or RPCs with request/response; pagination; idempotency keys for writes.
4. Data model	Entities, access patterns, SQL vs NoSQL choice, indexes, sharding key.
5. High-level design	Client, load balancer, stateless services, cache, database, queue, object storage, CDN.
6. Deep dives	The 1-2 hardest parts: hot keys, fan-out, ordering, consistency, failure handling.
7. Bottlenecks & trade-offs	Single points of failure, monitoring, what you would change at 10x scale.

High-level design building blocks

Block	Key points	Day
Load balancing Spread traffic over stateless servers	L4 vs L7; round robin, least connections, consistent hashing; health checks; sticky sessions only if forced	Day 24
Caching Serve hot reads from memory	Cache-aside, write-through, write-back; TTL + invalidation; LRU/LFU; thundering herd; Redis	Day 24
CDN Serve static/media from the edge	Push vs pull CDN; cache headers; signed URLs for private media	Day 24
CAP & consistency What you give up during a partition	CP vs AP; PACELC; strong, eventual, read-your-writes, monotonic reads	Day 24
API design Contracts between services	REST vs gRPC vs GraphQL vs WebSockets; status codes; cursor pagination; versioning; rate limits	Day 24
Auth & security Who you are and what you may do	Sessions vs JWT; OAuth 2.0; HTTPS/TLS; hashing passwords (bcrypt); least privilege	Day 24
Unique IDs IDs across many machines	UUID vs DB sequence vs Snowflake (time + machine + sequence); sortable IDs	Day 24
Databases Pick storage by access pattern	SQL (ACID, joins) vs key-value, document, wide-column, graph; isolation levels	Day 25
Indexes & storage engines Fast lookups on disk	B+Tree (read-optimised) vs LSM-tree (write-optimised); secondary indexes	Day 25

Block	Key points	Day
Replication Copies for availability and reads	Leader-follower, multi-leader, leaderless; sync vs async; quorum R + W > N; replica lag	Day 25
Sharding Split data across machines	Range vs hash partitioning; consistent hashing + virtual nodes; hot keys; rebalancing	Day 25
Object storage Big files and media	S3-style blob store + metadata in DB; multipart upload; lifecycle to cold storage	Day 25
Message queues & streams Decouple and absorb spikes	Kafka partitions, consumer groups, offsets; at-least-once + idempotent consumers; DLQ	Day 26
Consensus Agree on one leader / one log	Raft: terms, elections by majority, log replication; ZooKeeper/etcd for coordination	Day 26
Rate limiting Protect services from abuse	Token bucket, leaky bucket, fixed/sliding window; distributed with Redis + Lua	Day 26
Microservices Independent deployable services	API gateway, service discovery, circuit breaker, retries with backoff, bulkheads	Day 26
Distributed transactions Keep data correct across services	2PC vs saga; outbox pattern; CQRS & event sourcing	Day 26
Search Full-text queries	Inverted index, tokenisation, ranking; Elasticsearch; sync via change data capture	Day 26
Observability Know when and why it breaks	Logs, metrics, traces; SLIs/SLOs; p50/p99 latency; alerting	Day 26

Case studies to practise

Design	Key idea	Day
URL shortener	Base62 IDs from a key-generation service, read-heavy cache, 301 vs 302 redirects, analytics via queue	Day 27
Rate limiter (distributed)	Token bucket in Redis with Lua for atomicity, per-user and per-IP keys, 429 + Retry-After	Day 27
Chat app (WhatsApp)	WebSocket gateways, per-chat ordering with sequence numbers, offline inbox, delivery/read receipts	Day 27
News feed (Instagram/Twitter)	Fan-out on write for normal users, fan-out on read for celebrities, ranked feed cache	Day 27
Video streaming (YouTube)	Upload to blob store, async transcoding pipeline, adaptive bitrate, CDN delivery	Day 27
Ride hailing (Uber)	Driver locations in geohash/quadtree, matching service, trip state machine, surge pricing	Day 27

Design	Key idea	Day
Notification system	Priority queues per channel (push/SMS/email), templates, retries, user preferences, dedup	Day 27
Typeahead / autocomplete	Trie of top-K queries per prefix, offline rebuild from logs, edge caching	Day 27
Web crawler	URL frontier with politeness per domain, dedup with Bloom filter + content hash, robots.txt	Day 27
Google Drive / Dropbox	Chunked uploads, block-level dedup, metadata DB, sync via change notifications, conflict copies	Day 27
Ticket booking (BookMyShow)	Seat hold with TTL lock, payment saga, idempotent booking, handling flash traffic with a queue	Day 27
Distributed key-value store	Exactly your NexusKV project: LSM storage, consistent hashing, replication, leader election	Day 28

Low-level design

SOLID	Single responsibility, Open/closed, Liskov, Interface segregation, Dependency inversion	Day 22
Creational patterns	Singleton, Factory, Abstract Factory, Builder, Prototype	Day 22
Structural patterns	Adapter, Decorator, Proxy, Facade, Composite	Day 22
Behavioural patterns	Strategy, Observer, Command, State, Template Method, Chain of Responsibility	Day 22
Concurrency in Java	ExecutorService, locks, ConcurrentHashMap, BlockingQueue, CompletableFuture, deadlocks	Day 23
LLD case studies	Parking lot, elevator, BookMyShow, Splitwise, rate limiter class, LRU cache, logger, vending machine	Day 23

Numbers to know for estimates

L1 cache reference	`~1 ns`
Main memory reference	`~100 ns`
Read 1 MB from memory	`~250 us`
SSD random read	`~100 us`
Round trip in same datacenter	`~0.5 ms`
Read 1 MB from SSD	`~1 ms`
Disk seek (HDD)	`~10 ms`
Round trip India to US	`~150-250 ms`
1 million requests/day	`~12 QPS`
1 char / 1 int / 1 long	`1-2 B / 4 B / 8 B`

APPENDIX A

20 coding patterns: spot the pattern in 2 minutes

Pattern	Signal in the question	Classic problem
Two pointers	Sorted array, pair/triplet sums, in-place removal	3Sum
Sliding window	Longest/shortest contiguous subarray or substring with a condition	Minimum Window Substring
Fast & slow pointers	Cycles, middle of list, 'happy number'	Linked List Cycle II
Prefix sum + hash map	Count subarrays with a given sum/property	Subarray Sum Equals K
Cyclic sort	Numbers in range 1..n, find missing/duplicate	First Missing Positive
Merge intervals	Overlapping ranges, meetings, calendars	Merge Intervals
In-place list reversal	Reverse a list or a sub-part of it	Reverse Nodes in k-Group
Monotonic stack/deque	Next greater/smaller, spans, window max	Largest Rectangle in Histogram
Tree DFS / BFS	Paths, depths, level views, LCA	Binary Tree Maximum Path Sum
Graph BFS / DFS	Grids, islands, shortest unweighted path	Rotting Oranges
Topological sort	Dependencies, ordering, course prerequisites	Course Schedule II
Union-Find	Dynamic connectivity, grouping, redundant edges	Number of Provinces
Two heaps	Running median, balancing two halves	Find Median from Data Stream
Top-K / K-way merge	K largest/closest/frequent, merging sorted lists	Merge k Sorted Lists
Modified binary search	Sorted/rotated arrays, min/max feasible answer	Koko Eating Bananas
Backtracking	All subsets, permutations, combinations, boards	N-Queens
Dynamic programming	Count ways / min-max cost with overlapping choices	Edit Distance
Greedy	Locally best choice provably optimal (sort first)	Non-overlapping Intervals
Trie	Prefix search, autocomplete, many-word matching	Word Search II
Bit manipulation	XOR tricks, subsets as masks, n <= 20	Single Number

APPENDIX B

What the interview loops look like (typical, varies by team and level)

Google	Online assessment or phone screen, then about 4 to 5 interviews: mostly DSA (medium-hard, often graphs/DP), one Googleyness & leadership round, system design for L4+. Clean code and thinking aloud matter; a hiring committee decides.
Amazon	Online assessment (2 coding problems + work-style survey), then a loop of about 4 rounds: DSA, LLD/HLD for SDE-2, and Leadership Principles questions in every round. A Bar Raiser interviewer has veto power.
Microsoft	Online assessment or screen, then about 4 rounds: DSA, LLD/HLD (often a full design round for 3+ years), and an 'as appropriate' final round with a senior manager. Collaboration and growth mindset are probed.

PATTERN PLAYBOOK

How to approach each pattern, with a Java template

1. Two pointers

Use when: Sorted array, pair/triplet sums, in-place removal

1. Check the input is sorted (or can be sorted without losing what you need).
2. Place pointers: both ends for pair sums; same side (slow/fast) for in-place removal.
3. Compare the current pair with the target and move the pointer that brings you closer.
4. For triplets, fix one element in a loop and run two pointers on the rest; skip duplicates.

```
int l = 0, r = a.length - 1;
while (l < r) {
    int sum = a[l] + a[r];
    if (sum == target) return new int[]{l, r};
    if (sum < target) l++; else r--;
}
```

Common mistakes: Forgetting to skip duplicates in 3Sum; using two pointers on an unsorted array.
Complexity: O(n) after an O(n log n) sort; O(1) extra space.

Questions solved with this pattern (click to open on LeetCode):

Two Sum II - Input Array Is Sorted	Day 1	Medium
Container With Most Water	Day 2	Medium
3Sum	Day 2	Medium
Trapping Rain Water	Day 2	Hard
Sort Colors	Day 5	Medium
Valid Palindrome	Extra	Easy
Move Zeroes	Extra	Easy
Remove Duplicates from Sorted Array	Extra	Easy
4Sum	Extra	Medium

2. Sliding window

Use when: Longest/shortest contiguous subarray or substring with a condition

1. Confirm the answer is a contiguous subarray/substring.
2. Expand the right edge one step at a time and add the new element to the window state (sum or count map).
3. While the window breaks the rule, shrink from the left and remove that element from the state.
4. Update the answer at the point where the window is valid (longest: after shrinking; shortest: while shrinking).

```
Map<Character, Integer> cnt = new HashMap<>();
int l = 0, best = 0;
for (int r = 0; r < s.length(); r++) {
    cnt.merge(s.charAt(r), 1, Integer::sum);
    while (/* window invalid */ cnt.get(s.charAt(r)) > 1) {
        cnt.merge(s.charAt(l), -1, Integer::sum);
        l++;
    }
    best = Math.max(best, r - l + 1);
}
```

Common mistakes: Using sliding window with negative numbers (use prefix sums instead); updating the answer before the window is valid.
Complexity: O(n): each index enters and leaves the window once.

Questions solved with this pattern (click to open on LeetCode):

Best Time to Buy and Sell Stock	Day 2	Easy
Longest Substring Without Repeating Characters	Day 2	Medium
Minimum Window Substring	Day 2	Hard
Sliding Window Maximum	Day 7	Hard
Maximum Average Subarray I	Extra	Easy
Permutation in String	Extra	Medium
Longest Repeating Character Replacement	Extra	Medium
Find All Anagrams in a String	Extra	Medium
Minimum Size Subarray Sum	Extra	Medium

3. Fast & slow pointers

Use when: Cycles, middle of list, 'happy number'

1. Use when a linked list or a 'next value' function may contain a cycle, or you need the middle.
2. Move slow by 1 and fast by 2.
3. If they meet there is a cycle; reset one pointer to head and move both by 1 to find the cycle start.
4. For the middle, stop when fast reaches the end; slow is the middle.

```
ListNode slow = head, fast = head;
while (fast != null && fast.next != null) {
    slow = slow.next;
    fast = fast.next.next;
    if (slow == fast) {               // cycle found
        slow = head;
        while (slow != fast) { slow = slow.next; fast = fast.next; }
        return slow;                  // cycle start
    }
}
return null;
```

Common mistakes: Not checking fast.next before fast.next.next (NullPointerException).
Complexity: O(n) time, O(1) space.

Questions solved with this pattern (click to open on LeetCode):

Linked List Cycle	Day 6	Easy
Remove Nth Node From End of List	Day 6	Medium
Reorder List	Day 6	Medium
Middle of the Linked List	Extra	Easy
Happy Number	Extra	Easy
Palindrome Linked List	Extra	Easy
Linked List Cycle II	Extra	Medium
Find the Duplicate Number	Extra	Medium

4. Prefix sum + hash map

Use when: Count subarrays with a given sum/property

1. Look for 'number of subarrays with sum/xor/count = k' (works with negative numbers).
2. Keep a running prefix and a map from prefix value to how many times you have seen it.
3. At each index add map[prefix - k] to the answer, then record the current prefix.
4. Seed the map with prefix 0 seen once (the empty prefix).

```
Map<Integer, Integer> seen = new HashMap<>();
seen.put(0, 1);
int prefix = 0, count = 0;
for (int x : nums) {
    prefix += x;
    count += seen.getOrDefault(prefix - k, 0);
    seen.merge(prefix, 1, Integer::sum);
}
```

Common mistakes: Forgetting seen.put(0, 1); recording the prefix before reading the map.
Complexity: O(n) time and space.

Questions solved with this pattern (click to open on LeetCode):

Two Sum	Day 1	**Easy**
Product of Array Except Self	Day 2	**Medium**
Subarray Sum Equals K	Day 3	**Medium**
Range Sum Query 2D - Immutable	Day 19	**Medium**
Count of Range Sum	Day 19	**Hard**
Range Sum Query - Immutable	Extra	**Easy**
Contiguous Array	Extra	**Medium**
Continuous Subarray Sum	Extra	**Medium**
Subarray Sums Divisible by K	Extra	**Medium**

5. Cyclic sort

Use when: Numbers in range 1..n, find missing/duplicate

1. Signal: values are in the range 1..n (or 0..n) and you need missing or duplicate numbers.
2. Swap each value into its home index (value v belongs at index v - 1) until the current slot is correct or the swap would repeat.
3. Scan once more: any index i that does not hold i + 1 reveals a missing/duplicate number.

```
int i = 0;
while (i < n) {
    int home = a[i] - 1;
    if (a[i] > 0 && a[i] <= n && a[i] != a[home]) {
        int t = a[i]; a[i] = a[home]; a[home] = t;
    } else i++;
}
for (i = 0; i < n; i++) if (a[i] != i + 1) return i + 1;
return n + 1;
```

Common mistakes: Infinite loop when swapping duplicates (check a[i] != a[home], not i != home).
Complexity: O(n) time, O(1) space.

Questions solved with this pattern (click to open on LeetCode):

Missing Number	Day 2	**Easy**
Find All Duplicates in an Array	Day 2	**Medium**
First Missing Positive	Day 2	**Hard**

Find All Numbers Disappeared in an Array	Extra	Easy
Set Mismatch	Extra	Easy
Find the Duplicate Number	Extra	Medium

6. Merge intervals

Use when: Overlapping ranges, meetings, calendars

1. Sort intervals by start time.
2. Walk through them keeping the last merged interval.
3. If the next one overlaps (start <= last end), extend last end = max(last end, end); otherwise start a new one.
4. For 'maximum non-overlapping' sort by END instead and greedily keep the earliest-ending.

```
Arrays.sort(iv, (x, y) -> Integer.compare(x[0], y[0]));
List<int[]> out = new ArrayList<>();
for (int[] cur : iv) {
    if (out.isEmpty() || out.get(out.size() - 1)[1] < cur[0]) out.add(cur);
    else out.get(out.size() - 1)[1] = Math.max(out.get(out.size() - 1)[1], cur[1]);
}
return out.toArray(new int[0][]);
```

Common mistakes: Using a - b in comparators (overflow); deciding whether touching intervals count as overlapping.
Complexity: O(n log n) for the sort.

Questions solved with this pattern (click to open on LeetCode):

My Calendar I	Day 9	Medium
Merge Intervals	Day 15	Medium
Insert Interval	Day 15	Medium
Non-overlapping Intervals	Day 15	Medium
Minimum Number of Arrows to Burst Balloons	Day 15	Medium
Partition Labels	Day 15	Medium
My Calendar III	Day 19	Hard
Interval List Intersections	Extra	Medium
Car Pooling	Extra	Medium

7. In-place list reversal

Use when: Reverse a list or a sub-part of it

1. Use a dummy node if the head can change.
2. Walk to the node before the part to reverse.
3. Reverse with prev/cur/next pointers, then reconnect both ends.

```
ListNode prev = null, cur = head;
while (cur != null) {
    ListNode next = cur.next;
    cur.next = prev;
    prev = cur;
    cur = next;
}
return prev;
```

Common mistakes: Losing the rest of the list (save next before re-pointing); forgetting to reconnect the reversed part.
Complexity: O(n) time, O(1) space.

Questions solved with this pattern (click to open on LeetCode):

Reverse Linked List	Day 6	Easy
Reorder List	Day 6	Medium
Reverse Nodes in k-Group	Day 6	Hard
Palindrome Linked List	Extra	Easy
Reverse Linked List II	Extra	Medium
Swap Nodes in Pairs	Extra	Medium
Rotate List	Extra	Medium

8. Monotonic stack/deque

Use when: Next greater/smaller, spans, window max

1. Signal: 'next greater/smaller element', spans, or the max/min of every window.
2. Keep indices on a stack whose values are increasing or decreasing.
3. When the new element breaks the order, pop: each popped index has found its answer (the new element).
4. For window max, use a deque and also drop indices that slide out of the window from the front.

```
int[] ans = new int[n];
Deque<Integer> st = new ArrayDeque<>();   // indices, decreasing temps
for (int i = 0; i < n; i++) {
    while (!st.isEmpty() && t[i] > t[st.peek()]) {
        int j = st.pop();
        ans[j] = i - j;
    }
    st.push(i);
}
```

Common mistakes: Storing values instead of indices (you lose distances); choosing the wrong comparison for duplicates.
Complexity: O(n): each index is pushed and popped once.

Questions solved with this pattern (click to open on LeetCode):

Trapping Rain Water	Day 2	Hard
Valid Parentheses	Day 7	Easy
Daily Temperatures	Day 7	Medium
Min Stack	Day 7	Medium
Evaluate Reverse Polish Notation	Day 7	Medium
Largest Rectangle in Histogram	Day 7	Hard
Sliding Window Maximum	Day 7	Hard
Next Greater Element I	Extra	Easy
Next Greater Element II	Extra	Medium
Online Stock Span	Extra	Medium
Car Fleet	Extra	Medium
Remove K Digits	Extra	Medium
Sum of Subarray Minimums	Extra	Medium
Maximal Rectangle	Extra	Hard

9. Tree DFS / BFS

Use when: Paths, depths, level views, LCA

1. Decide what one call returns for a subtree (height, sum, is-valid, etc.).
2. Write the base case for null.
3. Combine left and right results; update a global answer if the best path can bend through the node.
4. Use BFS with queue.size() per level when the question says 'level', 'depth by depth' or 'right side view'.

```
int best = 0;
int depth(TreeNode node) {
    if (node == null) return 0;
    int l = depth(node.left), r = depth(node.right);
    best = Math.max(best, l + r);      // diameter through this node
    return 1 + Math.max(l, r);
}
```

Common mistakes: Mixing up what you return vs what you record globally; deep recursion on skewed trees (stack overflow).
Complexity: O(n) time; O(h) stack space.

Questions solved with this pattern (click to open on LeetCode):

Maximum Depth of Binary Tree	Day 8	**Easy**
Invert Binary Tree	Day 8	**Easy**
Diameter of Binary Tree	Day 8	**Easy**
Binary Tree Level Order Traversal	Day 8	**Medium**
Binary Tree Right Side View	Day 8	**Medium**
Lowest Common Ancestor of a Binary Tree	Day 8	**Medium**
Binary Tree Maximum Path Sum	Day 8	**Hard**
Serialize and Deserialize Binary Tree	Day 8	**Hard**
Convert Sorted Array to Binary Search Tree	Day 9	**Easy**
Validate Binary Search Tree	Day 9	**Medium**
Kth Smallest Element in a BST	Day 9	**Medium**
Lowest Common Ancestor of a Binary Search Tree	Day 9	**Medium**
Delete Node in a BST	Day 9	**Medium**
House Robber III	Day 18	**Medium**
Same Tree	Extra	**Easy**
Balanced Binary Tree	Extra	**Easy**
Subtree of Another Tree	Extra	**Easy**
Path Sum II	Extra	**Medium**
Binary Tree Zigzag Level Order Traversal	Extra	**Medium**
Construct Binary Tree from Preorder and Inorder Traversal	Extra	**Medium**

10. Graph BFS / DFS

Use when: Grids, islands, shortest unweighted path

1. Build the graph: adjacency list from edges, or treat each grid cell as a node.
2. Pick BFS for shortest path in steps; DFS for reachability, components or cycle checks.
3. Mark nodes visited when you push/enqueue them.
4. For several starting points (multi-source), enqueue all of them first.

```
Deque<int[]> q = new ArrayDeque<>();
q.offer(new int[]{sr, sc}); seen[sr][sc] = true;
int steps = 0;
int[][] dirs = {{1,0},{-1,0},{0,1},{0,-1}};
while (!q.isEmpty()) {
    for (int size = q.size(); size > 0; size--) {
        int[] c = q.poll();
        if (c[0] == tr && c[1] == tc) return steps;
        for (int[] d : dirs) {
            int r = c[0] + d[0], col = c[1] + d[1];
            if (r < 0 || col < 0 || r >= R || col >= C || seen[r][col] || grid[r][col] == '#')
 continue;
            seen[r][col] = true; q.offer(new int[]{r, col});
        }
    }
    steps++;
}
return -1;
```

Common mistakes: Marking visited on dequeue (nodes get added many times); forgetting disconnected components.
Complexity: O(V + E).

Questions solved with this pattern (click to open on LeetCode):

Number of Islands	Day 12	**Medium**
Clone Graph	Day 12	**Medium**
Rotting Oranges	Day 12	**Medium**
Pacific Atlantic Water Flow	Day 12	**Medium**
Surrounded Regions	Day 12	**Medium**
Open the Lock	Day 12	**Medium**
Word Ladder	Day 12	**Hard**
Shortest Path Visiting All Nodes	Day 18	**Hard**
Is Graph Bipartite?	Day 20	**Medium**
Bus Routes	Day 20	**Hard**
Reconstruct Itinerary	Day 20	**Hard**
Critical Connections in a Network	Day 20	**Hard**
Flood Fill	Extra	**Easy**
Max Area of Island	Extra	**Medium**
01 Matrix	Extra	**Medium**
Keys and Rooms	Extra	**Medium**
Shortest Path in Binary Matrix	Extra	**Medium**

11. Topological sort

Use when: Dependencies, ordering, course prerequisites

1. Signal: tasks with prerequisites, ordering, or 'detect a cycle in a directed graph'.
2. Build adjacency list and in-degree counts.
3. Queue every node with in-degree 0; pop, add to order, decrement neighbours' in-degree, enqueue those that hit 0.
4. If the order has fewer than n nodes, there is a cycle.

```
int[] indeg = new int[n];
List<List<Integer>> g = new ArrayList<>();
for (int i = 0; i < n; i++) g.add(new ArrayList<>());
for (int[] e : pre) { g.get(e[1]).add(e[0]); indeg[e[0]]++; }
Deque<Integer> q = new ArrayDeque<>();
for (int i = 0; i < n; i++) if (indeg[i] == 0) q.offer(i);
List<Integer> order = new ArrayList<>();
while (!q.isEmpty()) {
    int u = q.poll(); order.add(u);
    for (int v : g.get(u)) if (--indeg[v] == 0) q.offer(v);
}
return order.size() == n;   // false => cycle
```

Common mistakes: Reversing the edge direction for prerequisites [a, b] (b comes before a).
Complexity: O(V + E).

Questions solved with this pattern (click to open on LeetCode):

Course Schedule	Day 12	**Medium**
Course Schedule II	Day 12	**Medium**
Verifying an Alien Dictionary	Day 20	**Easy**
Longest Increasing Path in a Matrix	Day 30	**Hard**
Find Eventual Safe States	Extra	**Medium**
Minimum Height Trees	Extra	**Medium**
Course Schedule IV	Extra	**Medium**
Sort Items by Groups Respecting Dependencies	Extra	**Hard**

12. Union-Find

Use when: Dynamic connectivity, grouping, redundant edges

1. Signal: groups that merge over time, 'are these connected?', counting components, redundant edge.
2. Start with each node as its own parent.
3. union(a, b) joins the roots; find(x) follows parents with path compression.
4. If union finds both already in the same set, that edge creates a cycle.

```
int[] parent, rank;
int find(int x) { return parent[x] == x ? x : (parent[x] = find(parent[x])); }
boolean union(int a, int b) {
    int ra = find(a), rb = find(b);
    if (ra == rb) return false;
    if (rank[ra] < rank[rb]) { int t = ra; ra = rb; rb = t; }
    parent[rb] = ra;
    if (rank[ra] == rank[rb]) rank[ra]++;
    return true;
}
```

Common mistakes: Comparing parent[a] == parent[b] instead of find(a) == find(b).
Complexity: Almost O(1) per operation (inverse Ackermann).

Questions solved with this pattern (click to open on LeetCode):

Number of Islands	Day 12	Medium
Number of Provinces	Day 13	Medium
Redundant Connection	Day 13	Medium
Min Cost to Connect All Points	Day 13	Medium
Path With Minimum Effort	Day 13	Medium
Swim in Rising Water	Day 13	Hard
Number of Operations to Make Network Connected	Extra	Medium
Satisfiability of Equality Equations	Extra	Medium
Accounts Merge	Extra	Medium
Most Stones Removed with Same Row or Column	Extra	Medium

13. Two heaps

Use when: Running median, balancing two halves

1. Signal: median of a stream, or balancing a lower half and an upper half.
2. Max-heap holds the lower half, min-heap the upper half.
3. Add to the lower heap, move its top to the upper heap, then rebalance so lower has equal or one more element.
4. Median is lower.peek() (odd count) or the average of both tops.

```
PriorityQueue<Integer> lo = new PriorityQueue<>(Collections.reverseOrder());
PriorityQueue<Integer> hi = new PriorityQueue<>();
void add(int x) {
    lo.offer(x);
    hi.offer(lo.poll());
    if (hi.size() > lo.size()) lo.offer(hi.poll());
}
double median() {
    return lo.size() > hi.size() ? lo.peek() : (lo.peek() + (long) hi.peek()) / 2.0;
}
```

Common mistakes: Integer overflow when adding the two tops; forgetting to rebalance.
Complexity: O(log n) add, O(1) median.

Questions solved with this pattern (click to open on LeetCode):

Find Median from Data Stream	Day 10	Hard
Sliding Window Median	Extra	Hard
IPO	Extra	Hard

14. Top-K / K-way merge

Use when: K largest/closest/frequent, merging sorted lists

1. For 'K largest', keep a MIN-heap of size K; pop when it exceeds K.
2. For 'K smallest/closest', keep a MAX-heap of size K.
3. For merging K sorted lists, push the head of each list; pop the smallest and push its successor.

```
PriorityQueue<ListNode> pq = new PriorityQueue<>((a, b) -> Integer.compare(a.val, b.val));
for (ListNode h : lists) if (h != null) pq.offer(h);
ListNode dummy = new ListNode(0), tail = dummy;
while (!pq.isEmpty()) {
    ListNode n = pq.poll();
    tail.next = n; tail = n;
    if (n.next != null) pq.offer(n.next);
}
return dummy.next;
```

Common mistakes: Using the wrong heap direction; sorting everything (O(n log n)) when O(n log K) is expected.
Complexity: O(n log K).

Questions solved with this pattern (click to open on LeetCode):

Top K Frequent Elements	Day 3	**Medium**
Kth Largest Element in an Array	Day 5	**Medium**
Kth Largest Element in a Stream	Day 10	**Easy**
Last Stone Weight	Day 10	**Easy**
K Closest Points to Origin	Day 10	**Medium**
Task Scheduler	Day 10	**Medium**
Merge k Sorted Lists	Day 10	**Hard**
Network Delay Time	Day 13	**Medium**
Cheapest Flights Within K Stops	Day 13	**Medium**
Design Twitter	Day 14	**Medium**
Top K Frequent Words	Extra	**Medium**
Reorganize String	Extra	**Medium**
Kth Smallest Element in a Sorted Matrix	Extra	**Medium**
Find K Pairs with Smallest Sums	Extra	**Medium**
Smallest Range Covering Elements from K Lists	Extra	**Hard**

15. Modified binary search

Use when: Sorted/rotated arrays, min/max feasible answer

1. Signal: sorted/rotated data, or 'minimum X such that condition holds' where the condition is monotonic.
2. Define the search range [lo, hi] of possible answers.
3. Write a check(mid) that is false...false, true...true across the range.
4. Use the lower-bound template so the loop ends exactly at the first true.

```
int lo = 1, hi = maxPile;           // answer range
while (lo < hi) {
    int mid = lo + (hi - lo) / 2;
    if (canFinish(mid)) hi = mid;      // mid works, try smaller
    else lo = mid + 1;
}
return lo;
```

Common mistakes: Infinite loops from mixing lo <= hi with hi = mid; overflow in (lo + hi) / 2.

Complexity: O(log range x cost of check).

Questions solved with this pattern (click to open on LeetCode):

Two Sum II - Input Array Is Sorted	Day 1	Medium
Binary Search	Day 5	Easy
Search in Rotated Sorted Array	Day 5	Medium
Find Minimum in Rotated Sorted Array	Day 5	Medium
Koko Eating Bananas	Day 5	Medium
Search a 2D Matrix	Day 5	Medium
Median of Two Sorted Arrays	Day 5	Hard
Time Based Key-Value Store	Day 14	Medium
Snapshot Array	Day 14	Medium
Random Pick with Weight	Day 18	Medium
Search Insert Position	Extra	Easy
Find First and Last Position of Element in Sorted Array	Extra	Medium
Find Peak Element	Extra	Medium
Capacity To Ship Packages Within D Days	Extra	Medium
Split Array Largest Sum	Extra	Hard

16. Backtracking

Use when: All subsets, permutations, combinations, boards

1. Draw the decision tree: what choice is made at each level?
2. Recurse with the partial path; when it is complete, copy it into the result.
3. Undo the last choice after returning (backtrack).
4. Prune branches that can no longer lead to a valid answer.

```
void dfs(int start, List<Integer> path) {
    res.add(new ArrayList<>(path));            // subsets: record every node
    for (int i = start; i < nums.length; i++) {
        if (i > start && nums[i] == nums[i - 1]) continue;   // skip dupes (sorted)
        path.add(nums[i]);
        dfs(i + 1, path);
        path.remove(path.size() - 1);          // undo
    }
}
```

Common mistakes: Adding the path itself instead of a copy; forgetting to undo the choice.
Complexity: Exponential: O(2^n) subsets, O(n!) permutations.

Questions solved with this pattern (click to open on LeetCode):

Subsets	Day 4	Medium
Permutations	Day 4	Medium
Combination Sum	Day 4	Medium
Generate Parentheses	Day 4	Medium
Word Search	Day 4	Medium
N-Queens	Day 4	Hard
Word Search II	Day 11	Hard
Letter Combinations of a Phone Number	Extra	Medium
Subsets II	Extra	Medium

Permutations II	Extra	Medium
Combination Sum II	Extra	Medium
Palindrome Partitioning	Extra	Medium
Sudoku Solver	Extra	Hard

17. Dynamic programming

Use when: Count ways / min-max cost with overlapping choices

1. Write a brute-force recursion that tries every choice.
2. Identify the state: the minimum arguments that describe a subproblem (index, remaining capacity, ...).
3. Write the transition (how a state uses smaller states) and the base cases.
4. Add memoisation; then convert to a bottom-up table if needed and shrink space.

```
// Coin change: fewest coins to make amount
int[] dp = new int[amount + 1];
Arrays.fill(dp, amount + 1);
dp[0] = 0;
for (int a = 1; a <= amount; a++)
    for (int c : coins)
        if (c <= a) dp[a] = Math.min(dp[a], dp[a - c] + 1);
return dp[amount] > amount ? -1 : dp[amount];
```

Common mistakes: Wrong iteration order (0/1 knapsack must loop capacity backwards); using Integer.MAX_VALUE then adding 1 (overflow).
Complexity: Number of states x cost per transition.

Questions solved with this pattern (click to open on LeetCode):

Maximum Subarray	Day 2	Medium
Longest Palindromic Substring	Day 11	Medium
Climbing Stairs	Day 16	Easy
House Robber	Day 16	Medium
House Robber II	Day 16	Medium
Decode Ways	Day 16	Medium
Coin Change	Day 16	Medium
Partition Equal Subset Sum	Day 16	Medium
Longest Increasing Subsequence	Day 16	Medium
Word Break	Day 16	Medium
Unique Paths	Day 17	Medium
Minimum Path Sum	Day 17	Medium
Longest Common Subsequence	Day 17	Medium
Edit Distance	Day 17	Medium
Palindromic Substrings	Day 17	Medium
Best Time to Buy and Sell Stock with Cooldown	Day 17	Medium
Distinct Subsequences	Day 17	Hard
Regular Expression Matching	Day 17	Hard
Counting Bits	Day 18	Easy
House Robber III	Day 18	Medium
Target Sum	Day 18	Medium

Burst Balloons	Day 18	**Hard**
Min Cost Climbing Stairs	Extra	**Easy**
Coin Change II	Extra	**Medium**
Maximal Square	Extra	**Medium**
Longest Palindromic Subsequence	Extra	**Medium**
Interleaving String	Extra	**Medium**

18. Greedy

Use when: Locally best choice provably optimal (sort first)

1. Find a natural order (sort by end time, by ratio, by deadline).
2. Make the locally best choice and never undo it.
3. Convince yourself with an exchange argument; if a counter-example exists, switch to DP.

```
Arrays.sort(iv, (x, y) -> Integer.compare(x[1], y[1]));   // by end
int kept = 0, end = Integer.MIN_VALUE;
for (int[] cur : iv) {
    if (cur[0] >= end) { kept++; end = cur[1]; }
}
return iv.length - kept;   // removals for non-overlapping
```

Common mistakes: Assuming greedy works without checking a counter-example (e.g. coins {1, 3, 4} for 6).
Complexity: Usually O(n log n) for the sort.

Questions solved with this pattern (click to open on LeetCode):

Best Time to Buy and Sell Stock	Day 2	**Easy**
Maximum Subarray	Day 2	**Medium**
Task Scheduler	Day 10	**Medium**
Jump Game	Day 15	**Medium**
Gas Station	Day 15	**Medium**
Partition Labels	Day 15	**Medium**
Non-overlapping Intervals	Day 15	**Medium**
Minimum Number of Arrows to Burst Balloons	Day 15	**Medium**
Assign Cookies	Extra	**Easy**
Jump Game II	Extra	**Medium**
Boats to Save People	Extra	**Medium**
Hand of Straights	Extra	**Medium**
Valid Parenthesis String	Extra	**Medium**
Candy	Extra	**Hard**

19. Trie

Use when: Prefix search, autocomplete, many-word matching

1. Signal: many words share prefixes; prefix search; autocomplete; searching many words in a grid.
2. Each node has children[26] and an end-of-word flag (or the word itself).
3. Insert and search walk one character per level.
4. For grid word search, DFS on the board while walking the trie; prune finished branches.

```
class Node { Node[] next = new Node[26]; boolean end; }
Node root = new Node();
void insert(String w) {
    Node cur = root;
    for (char ch : w.toCharArray()) {
        int i = ch - 'a';
        if (cur.next[i] == null) cur.next[i] = new Node();
        cur = cur.next[i];
    }
    cur.end = true;
}
```

Common mistakes: Returning true for a prefix when a full word was asked (check the end flag).
Complexity: O(length of word) per operation.

Questions solved with this pattern (click to open on LeetCode):

Implement Trie (Prefix Tree)	Day 11	**Medium**
Design Add and Search Words Data Structure	Day 11	**Medium**
Word Search II	Day 11	**Hard**
Search Suggestions System	Day 27	**Medium**
Replace Words	Extra	**Medium**
Longest Word in Dictionary	Extra	**Medium**
Map Sum Pairs	Extra	**Medium**
Maximum XOR of Two Numbers in an Array	Extra	**Medium**

20. Bit manipulation

Use when: XOR tricks, subsets as masks, n <= 20

1. Signal: 'appears once while others appear twice', subsets with n <= 20, powers of two, flags.
2. Remember the core tricks: a ^ a = 0, x & (x - 1) clears the lowest set bit, x & -x isolates it.
3. Use a mask from 0 to (1 << n) - 1 to enumerate every subset.

```
for (int mask = 0; mask < (1 << n); mask++) {
    List<Integer> subset = new ArrayList<>();
    for (int i = 0; i < n; i++)
        if ((mask & (1 << i)) != 0) subset.add(nums[i]);
}
int single = 0; for (int x : nums) single ^= x;
```

Common mistakes: Operator precedence: always bracket (mask & (1 << i)) != 0; >> vs >>> for negative numbers.
Complexity: O(2^n x n) for subset enumeration.

Questions solved with this pattern (click to open on LeetCode):

Missing Number	Day 2	**Easy**
Subsets	Day 4	**Medium**
Single Number	Day 18	**Easy**
Counting Bits	Day 18	**Easy**

Number of 1 Bits	Day 18	**Easy**
Shortest Path Visiting All Nodes	Day 18	**Hard**
Power of Two	Extra	**Easy**
Reverse Bits	Extra	**Easy**
Single Number II	Extra	**Medium**
Sum of Two Integers	Extra	**Medium**
Bitwise AND of Numbers Range	Extra	**Medium**

APPENDIX C

Java DSA syntax cheat sheet

Stack / Queue

```
Deque<Integer> st = new ArrayDeque<>(); st.push(x); st.pop(); st.peek();
Queue<Integer> q = new ArrayDeque<>(); q.offer(x); q.poll();
```

Heaps

```
PriorityQueue<Integer> min = new PriorityQueue<>();
PriorityQueue<Integer> max = new PriorityQueue<>(Collections.reverseOrder());
PriorityQueue<int[]> pq = new PriorityQueue<>((a, b) -> a[1] - b[1]);
```

Maps

```
map.merge(k, 1, Integer::sum); map.getOrDefault(k, 0);
map.computeIfAbsent(k, x -> new ArrayList<>()).add(v);
```

TreeMap

```
tm.floorKey(x); tm.ceilingKey(x); tm.firstKey(); tm.pollFirstEntry(); tm.headMap(x);
```

Sorting

```
Arrays.sort(a); Arrays.sort(iv, (x, y) -> Integer.compare(x[0], y[0]));
list.sort(Comparator.comparing(Person::age).thenComparing(Person::name));
```

Strings

```
char[] c = s.toCharArray(); new String(c); String.valueOf(c);
StringBuilder sb = new StringBuilder(); sb.append(x).reverse().toString();
```

Arrays

```
int[] a = new int[n]; Arrays.fill(a, -1); int[][] g = new int[r][c];
int[] copy = Arrays.copyOfRange(a, l, r); Arrays.toString(a);
```

Overflow & math

```
long x = (long) a * b; Integer.MAX_VALUE; Math.floorMod(a, m);
int mid = lo + (hi - lo) / 2; Integer.bitCount(x);
```

APPENDIX D

Revision system

- Spaced repetition: re-solve each problem you needed help on after 2 days, 7 days and 21 days.
- Every Sunday (Days 7, 14, 21, 28): 30 minutes reading all sticky notes of the week.
- Keep a mistake log (problem, what tripped you, the fix) in the GitHub repo.
- Days 29-30: read all 30 sticky notes and the patterns table once more.